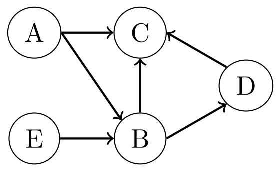
Worksheet 1

Exercise 1: Graph terminology

Have a look at the following causal graphical model.



a) Is the represented graphical model a directed acyclic graph (DAG)? Justify your answer.

b) Give 3 examples of vertices-sequences that correspond to a directed path and undirected paths.

c) Determine children, parents and ancestors for the two nodes B and D.

d) Is the total cause of B on C transported by the path B $\rightarrow$ C? Justify your answer.

e) Draw the graph in R using the function `dag` from the R-Package `gRbase` and repeat c) by using R-Commands.

Exercise 2: Travel time

A new inter-city route A was built between two cities. In order to compare the new route A with the old route B, the travel time in minutes was measured for each route by 250 randomly selected drivers. The data set `reisezeit` in the file `NewRoute.rda` contains the measured times.

a) Analyze the data and evaluate whether the travel time has been reduced with the new route A or not (time route A $<$ time route B).

b) Draw a causal diagram visualizing the relation between Time, Route and Weekday.

c) What interventions would need to be taken to estimate the causal effect of the route on travel time?

d) How would you design experimental data to measure the effect of the new Route?

Exercise 3: Weightlifting

A study aims to analyze the effects of weight and technique on weightlifting performance, which is measured by the number of repetitions. The researchers hypothesize that men tend to select heavier weights due to overconfidence. Additionally, men are generally stronger and therefore able to perform more repetitions. The chosen weight is also believed to influence lifting technique.

a) Draw a causal diagram of the four variables `technique`, `weight`, `gender` and `performance`.

b) Which intervention would need to be carried out to estimate a causal effect of weight on performance?

c) How does the causal graphical model look like under the intervention b)?

d) Does the causal effect of weight on performance change when all participants receive a specific instruction on the technique?

Exercise 4: Oberservational vs. Experimental Data

a) You recruit 300 persons suffering from cold symptoms. You feed all 300 persons all lots of oranges, which are full of vitamin C. Eight days later, 292 out of 300 people are free of cold symptoms. Did the oranges cure people?

b) You recruit 300 persons for a winter race on a test track. You assign randomly 150 persons to cars with winter tires and the other 150 persons to cars with summer tires. There are 30 fewer breakdowns in the group with winter tires. Is driving with winter tires safer?

c) You are comparing two female teams of floorball players: U21 vs. seniors. There were 8 cases of cruciate ligament injuries in the U21 team last year and only 1 case in the senior team. Can we conclude from this that the senior team is stronger?

Exercise 5: Pearl's ladder

Formulate possible research questions on association, intervention and imagination for the following pairs (with reference to Pearl's ladder).

a) Effect of snow and ice on car accidents

b) Effect of smoking on lung cancer

c) Effect of junk food on obesity

d) Effect of wind on speed of sailboat